

UNIVERSIDAD TECNOLÓGICA NACIONAL

TECNICATURA EN PROGRAMACIÓN

LA APLICACIÓN EN INTERNET

Trabajo Práctico 1

Consigna de trabajo práctico · Unidad de Despliegue en un VPS

2Prog3, Programación III

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Dominio: matiaslimina.me

Datos del despliegue

URL del frontend: <https://calculadora.matiaslimina.me/>

URL de la API: <https://calculadora.api.limina.me/>

Proveedor VPS: Hostinger

Proveedor Dominio: Namecheap

R1 — Dominio propio con resolución correcta

Registro del grupo: <https://calculadora.matiaslimina.me/>

R2 — VPS asegurada

Verificación: un barrido de puertos desde afuera no debe mostrar nada más que esos tres.

Fundamento: Clase 2, secciones 2.5 a 2.9.

Inicio de activación de Firewall:

Se entra al servidor con la clave ssh.

Establecer políticas predeterminadas (principio de valores predeterminados seguros):

```
sudo ufw default deny incoming
```

```
sudo ufw default allow outgoing
```

Abrir los únicos puertos necesarios:

```
sudo ufw allow 22/tcp
```

```
sudo ufw allow 80/tcp
```

```
sudo ufw allow 443/tcp
```

(El puerto 80 debe estar abierto obligatoriamente para que Let's Encrypt pueda validar y renovar los certificados SSL).

Levantar firewall:

```
sudo ufw enable
```

Se verifica el estado (activo):

```
sudo ufw status verbose
```

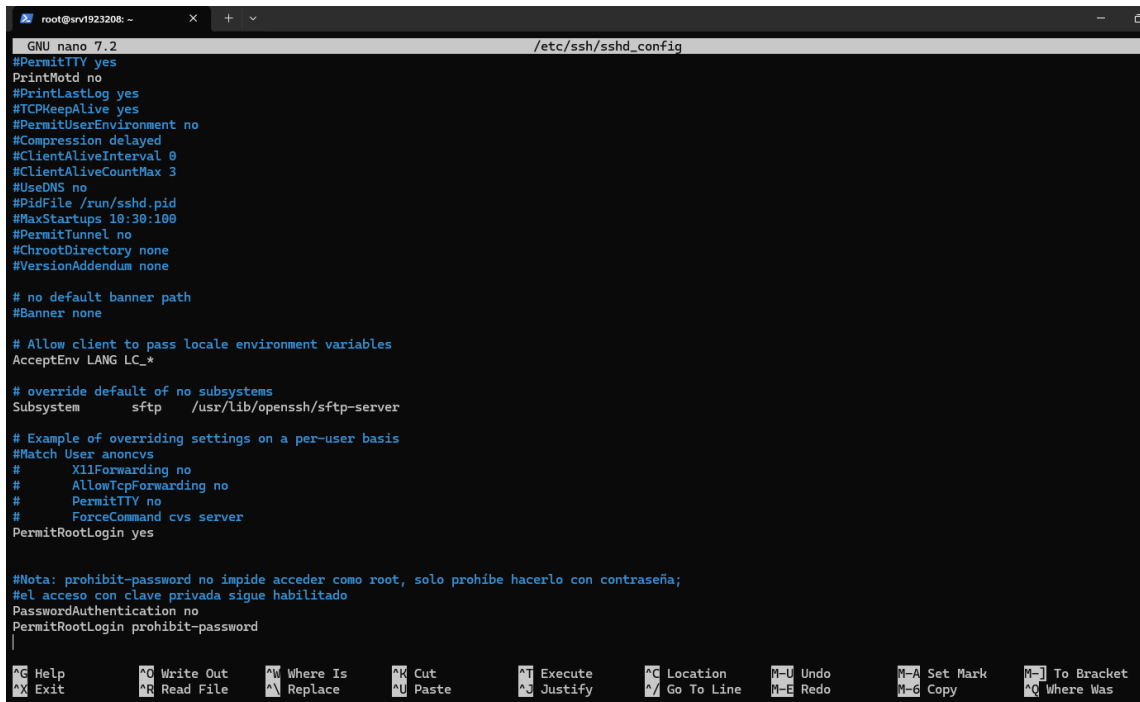
Se modifica el "sshd_config" agregando los comandos para impedir el acceso por clave:

```
sudo nano /etc/ssh/sshd_config
```

```
PasswordAuthentication no
```

```
PermitRootLogin prohibit-password
```

Nota: prohibit-password no impide acceder como root, solo prohíbe hacerlo con contraseña; el acceso con clave privada sigue habilitado



```
root@srv1923208: ~  
GNU nano 7.2 /etc/ssh/sshd_config  
#PermitTTY yes  
PrintMotd no  
#PrintLastLog yes  
#TCPKeepAlive yes  
#PermitUserEnvironment no  
#Compression delayed  
#ClientAliveInterval 0  
#ClientAliveCountMax 3  
#UseDNS no  
#PidFile /run/sshd.pid  
#MaxStartups 10:30:100  
#PermitTunnel no  
#ChrootDirectory none  
#VersionAddendum none  
  
# no default banner path  
#Banner none  
  
# Allow client to pass locale environment variables  
AcceptEnv LANG LC_*  
  
# override default of no subsystems  
Subsystem sftp /usr/lib/openssh/sftp-server  
  
# Example of overriding settings on a per-user basis  
#Match User anoncvs  
#    X11Forwarding no  
#    AllowTcpForwarding no  
#    PermitTTY no  
#    ForceCommand cvs server  
PermitRootLogin yes  
  
#Nota: prohibir-password no impide acceder como root, solo prohíbe hacerlo con contraseña;  
#el acceso con clave privada sigue habilitado  
PasswordAuthentication no  
PermitRootLogin prohibit-password  
  
^G Help      ^O Write Out  ^M Where Is   ^X Cut        ^T Execute    ^C Location   ^U Undo       ^_ Set Mark   ^- To Bracket  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^_ Justify    ^/ Go To Line  ^-E Redo     ^-C Copy     ^Q Where Was
```

Se recarga el sistema:
sudo systemctl reload ssh

```

root@srv1923208: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\Aida\.ssh> ssh root@179.199.130.181
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep 21 21:45:10 UTC 2026

System load:  0.0          Processes:      153
Usage of /:   11.4% of 47.39GB Users logged in: 1
Memory usage: 13%         IPv4 address for eth0: 179.199.130.181
Swap usage:   0%          IPv6 address for eth0: 2a02:4780:6e:afe9::1

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log

Last login: Mon Sep 21 21:38:14 2026 from 169.254.0.1
root@srv1923208:~# sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@srv1923208:~# sudo ufw status verbose
Status: active

```

```

root@srv1923208: ~
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log

Last login: Mon Sep 21 21:38:14 2026 from 169.254.0.1
root@srv1923208:~# sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@srv1923208:~# sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
---
22/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
443/tcp (v6) ALLOW IN Anywhere (v6)

root@srv1923208:~# sudo nano /etc/ssh/sshd_config
root@srv1923208:~# sudo nano /etc/ssh/sshd_config
root@srv1923208:~# sudo systemctl reload ssh
root@srv1923208:~# |

```

En otra terminal se ejecuta este comando para verificar que no se puede entrar sin ssh:
ssh -o PubkeyAuthentication=no root@179.199.130.181

```
root@srv1923208: ~
PS C:\Users\Aida\.ssh> ssh root@179.199.130.181
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-139-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Mon Sep 21 21:58:18 UTC 2026

System load:  0.05          Processes:            157
Usage of /:   11.4% of 47.39GB Users logged in:      1
Memory usage: 13%          IPv4 address for eth0: 179.199.130.181
Swap usage:   0%           IPv6 address for eth0: 2a02:4780:6e:afe9::1

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https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log

Last login: Mon Sep 21 21:57:24 2026 from 104.28.214.170
root@srv1923208:~# ssh -o PubkeyAuthentication=no root@179.199.130.181
The authenticity of host '179.199.130.181 (179.199.130.181)' can't be established.
ED25519 key fingerprint is SHA256:vQ0ffCm7GUH1iboRa+1SCIfuCXxPGLCWTeS/aCqUD+U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '179.199.130.181' (ED25519) to the list of known hosts.
root@179.199.130.181's password: |
```

R3 — Imágenes construidas desde el repositorio

Logs de construcción (backend):

```
#11 exporting layers
#11 exporting layers 3.0s done
#11 exporting manifest sha256:01be2f9a12b90323c2739865082dd060099e3d44e127128b15939559ee1c8151
#11 exporting manifest sha256:01be2f9a12b90323c2739865082dd060099e3d44e127128b15939559ee1c8151 done
#11 exporting config sha256:21f46f223c4f7895ba7d6d57ee9aec8c57351bbd1021c6b24afc99a41dde6ade done
#11 exporting attestation manifest sha256:c0743d97b3e2e97d499395b385c51f516f502ce00ce3ebac26ad52ba153979c0 0.0s done
#11 exporting manifest list sha256:7966b6606ce94a66913893fde8a97b7a28bc3767caa32425aa6fa2a3df90f082 0.0s done
#11 naming to docker.io/easypanel/gordos_programacion/calc-back:latest done
#11 unpacking to docker.io/easypanel/gordos_programacion/calc-back:latest
#11 unpacking to docker.io/easypanel/gordos_programacion/calc-back:latest 1.9s done
#11 DONE 5.0s
#####
### Success
### Fri, 18 Sep 2026 14:46:46 GMT
#####
```

Logs de construcción (frontend):

```
#9 exporting layers 0.3s done
#9 exporting manifest sha256:ae10f2efd4c7ef2d681a1431588bab7856edbc0683bedb0c2454700ae3b3df71 0.0s done
#9 exporting config sha256:75958b4a55044b114e35a23298a95f2ed6b50b442feaae753ac253737885641e 0.0s done
#9 exporting attestation manifest sha256:4c64f9696fcc0825e97c4d510b3eb2bd4507fc37b1de7a5d7cc8b8e0c0855fc3 0.0s done
#9 exporting manifest list sha256:066669283e26e3e576cea84a6a47936b12fc5b295a53f23b54f0c18d414e148b
#9 exporting manifest list sha256:066669283e26e3e576cea84a6a47936b12fc5b295a53f23b54f0c18d414e148b 0.0s done
#9 naming to docker.io/easypanel/gordos_programacion/calc-front:latest done
#9 unpacking to docker.io/easypanel/gordos_programacion/calc-front:latest 0.1s done
#9 DONE 0.6s
#####
### Success
### Fri, 18 Sep 2026 14:46:27 GMT
#####
```

R4 — Publicación con HTTPS válido

curl.exe -I https://calculadora.matiaslimina.me/

```
PS C:\Users\matia> curl.exe -I https://calculadora.matiaslimina.me/
HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Length: 34245
Content-Type: text/html
Date: Sat, 19 Sep 2026 20:38:49 GMT
Etag: "6aad4bd6-85c5"
Last-Modified: Fri, 18 Sep 2026 14:33:58 GMT
Server: nginx/1.31.6
```

curl.exe -I http://calculadora.matiaslimina.me/

```
PS C:\Users\matia> curl.exe -I http://calculadora.matiaslimina.me/
HTTP/1.1 308 Permanent Redirect
Location: https://calculadora.matiaslimina.me/
Date: Sat, 19 Sep 2026 20:40:02 GMT
Content-Length: 18
```

R5 — El contrato funciona de punta a punta
Prueba de funcionamiento correcto:

The screenshot shows a web application titled "Calculadora" with a dark theme. It has two input fields for "PRIMER VALOR" and "SEGUNDO VALOR", both containing the number "5". Below these are buttons for operations: "+ SUMA", "-", "x MULTIPLICACIÓN", "÷ DIVISIÓN", and "LOG₁₀ LOGARITMO". A large blue button "Enviar a la API" and a grey button "Limpiar" are at the bottom of the input section. The result area shows "5 + 5 =" followed by a large green "10".

To the right, the Chrome DevTools Network tab is open, showing two requests:

Status	Meth...	Domain	File	Initiator	Type	Transferred	Size
200	POST	api.matiasli...	calcular	¿573 (fetch)	json	319 B	98 B
200	OPTI...	api.matiasli...	calcular	fetch	plain	397 B	2 B

At the bottom of the network tab, it shows "2 requests", "100 B / 716 B transferred", and "Finish: 63 ms".

Prueba en caso de division por cero:

Calculadora

Este front no calcula nada. Le manda los datos a una API que corre en **otro servidor, en otro puerto**, y muestra lo que le responde. Hoy esa API está hecha en FastAPI — pero a esta página no le consta.

PRIMER VALOR: 5 SEGUNDO VALOR: 0

OPERACIÓN: + SUMA, − RESTA, × MULTIPLICACIÓN, ÷ DIVISIÓN, LOG₁₀ LOGARITMO

Enviar a la API Limpiar

No se puede dividir por cero.

► Ver el intercambio HTTP crudo

Status	Meth...	Domain	File	Initiator	Type	Transferred	Size
200	POST	api.matiassi...	calcular	/575 (fetch)	json	319 B	98 B
200	OPTI...	api.matiassi...	calcular	fetch	plain	397 B	2 B
400	POST	api.matiassi...	calcular	/575 (fetch)	json	263 B	42 B

3 requests 142 B / 979 B transferred Finish: 2.33 min

R6 — Persistencia en base de datos aislada

Prueba de persistencia:

ÚLTIMAS OPERACIONES Actualizar

Las últimas 6 operaciones, guardadas en una base de datos que no está publicada en internet.

5.0 * 4.0 = 20.0	19/9, 17:51
5.0 + 4.0 = 9.0	19/9, 17:51
5.0 - 4.0 = 1.0	19/9, 17:51
5.0 / 4.0 = 1.25	19/9, 17:51
7.0 - 1.0 = 6.0	18/9, 11:47
log ₁₀ (60.0) = 1.7781512503836436	18/9, 11:47

Prueba de conexion cerrada:

```
PS C:\Users\matia> nmap -p 5432 179.199.130.181
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-19 18:00 -0300
Nmap scan report for srv1923208.hstgr.cloud (179.199.130.181)
Host is up (0.048s latency).

PORT      STATE      SERVICE
5432/tcp  filtered  postgresql

Nmap done: 1 IP address (1 host up) scanned in 1.05 seconds
PS C:\Users\matia> |
```

R7 — Degradación elegante

Operaciones sin persistencia:

PRIMER VALOR: 5

SEGUNDO VALOR: 4

OPERACIÓN:

+ SUMA - RESTA **× MULTIPLICACIÓN** ÷ DIVISIÓN LOG₁₀ LOGARITMO

Enviar a la API Limpiar

5 * 4 =

20

► Ver el intercambio HTTP crudo

PRIMER VALOR

5

SEGUNDO VALOR

4

OPERACIÓN

+

SUMA

−

RESTA

×

MULTIPLICACIÓN

÷

DIVISIÓN

LOG₁₀

LOGARITMO

Enviar a la API

Limpiar

5 - 4 =

1

► Ver el intercambio HTTP crudo

ÚLTIMAS OPERACIONES

Esta API está corriendo sin base de datos, así que no guarda historial. La calculadora funciona

Operaciones con persistencia:

5

4

OPERACIÓN

+

SUMA

−

RESTA

×

MULTIPLICACIÓN

÷

DIVISIÓN

LOG₁₀

LOGARITMO

Enviar a la API

Limpiar

5 + 4 =

9

► Ver el intercambio HTTP crudo

ÚLTIMAS OPERACIONES

Actualizar

Las últimas 8 operaciones, guardadas en una base de datos que no está publicada en internet.

5.0 + 4.0 = 9.0

19/9, 18:06

Problemas encontrados:

Un compañero no podía conectarse

Solucion a traves de un tracert pudimos identificar que se trataba de un problema de su proveedor y lo solucionamos a traves de un vpn

La base de datos se desconecto al agregar una nueva feature

Una vez identificado el problema simplemente tuvimos que actualizar la URL dentro de las variables del backend